Colby T. Ford, Ph.D.

Cloud Data, AI, & Genomics Architect,
Computational Biologist, & Data Scientist

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Education

The University of North Carolina at Charlotte

2018 Doctor of Philosophy in Bioinformatics and Computational Biology
Advisor: Daniel Janies, Ph.D.
Wayland H. Cato Doctoral Fellow

2015 Master of Science in Data Science and Business Analytics
Advisor: Mirsad Hadžikadić, Ph.D.

2014 Bachelor of Arts in Applied Mathematics, Concentration in Statistics
Advisor: Mary Kim Harris, Ed.D.
Coffey Scholar, C.L. Robbins Scholar

Industry Positions

Tuple *(a Microsoft Partner)*

Founded Tuple, a Microsoft and Databricks partner firm specializing in scalable cloud -omics solutions. Tuple provides data architecture and research consulting services in Azure, focusing on cloud AI in the life sciences. Co-wrote the Azure Data Scientist Microsoft Certified Solution Associate exam (DP-100), the Azure MLOps/AIOps exam (AI-300), the Designing and Implementing Multi-Agent AI Solutions exam (AI-500), the Azure Databricks Engineering exam (DP-750), and the MS Learn Azure Data Scientist exam. Created the `azbasespace` connector for Illumina BaseSpace to Azure and the `ahab` system for -omics in Kubernetes. Microsoft Partner-of-the-Year nominee for 2023 and 2026.

August 2019 - Present Owner & Principal Consultant

Silico Biosciences *(a subsidiary of Tuple)*

Founded Silico Biosciences, an AI-driven drug discovery company. Silico uses a scalable computational framework to generate antibody candidates using custom-trained AI models and evaluate them *in silico*. Currently, Silico Biosciences has created antibody candidates for oncotargets PD-1 and EGFR, and viral targets including H5N1 influenza, Nipah virus, and Ebola.

May 2024 - Present Founder & Computational Structural Biologist

Axia Medicine

Supporting Axia Medicine, a precision medicine startup, in the development of their *smartomics* platform. This platform enables patients, providers, and pharmaceutical companies to build clinical cohorts with ease. Axia provides the infrastructure to broker the collection of multi-omics data, lab results, images, and metadata to build custom, targeted cohorts for research.

Nov 2025 - Present Co-Founder & Fractional CTO

Pfizer

Established a scalable cloud computing architecture on Azure to support the Translational Sciences group in the discovery of therapeutic targets for antibody development in oncology. This includes an automated, Kubernetes-based workflow for scalable RNA-seq, WES, protein docking, and pharmacodynamics pipelines along with a supporting GxP-qualified data platform, data lake, and machine learning architecture. Collaborated with pre-clinical and clinical teams to design scalable, cloud-based protein structural analyses for antibody candidates against tumor targets. Trained and productionalized large drug sensitivity ML models using gene expression data that were used in patient selection in clinical trials.

December 2023 - May 2024	Mgr. ML & Cloud Engineering, Bioinformatics, Pfizer Oncology
September 2021 - December 2023	Mgr. ML & Cloud Engineering, Bioinformatics, Seagen
March 2021 - September 2021	Sr. Scientist, Bioinformatics, Seagen

Amissa Health

Co-founded Amissa, Inc., a digital health company that uses cloud-based AI and analytics through consumer wearable devices in the fight against Alzheimer's disease and other conditions. Designed and built the *Sinuo* cloud platform, which applies state-of-the-art artificial intelligence techniques to sensory data, digital biomarkers, and additional patient metrics. Amissa's work was funded by the NIH National Institute on Aging SBIR program (>$3.5M), NC state SBIR funds, and the NSF I-Corps program.

March 2020 - October 2024 Co-Founder & VP of Technology

BlueGranite *(a Microsoft Partner)*

Developed data platform and AI solutions using the Microsoft Azure cloud with services such as Azure Machine Learning, Cognitive Services, and Databricks (Apache Spark). Hosted many training workshops and gave conference presentations and demonstrations on Microsoft advanced analytics technologies. Started and led the genomics consulting practice, building scalable solutions for bioinformatics analyses in Azure. (Acquired by 3Cloud, LLC.)

May 2020 - October 2021	Principal Architect, Life Sciences
December 2020 - May 2020	Genomics Practice Lead
December 2020 - May 2020	AI Practice Lead (Interim)
November 2018 - December 2020	Solution Architect, Artificial Intelligence
January 2017 - November 2018	Senior Data Scientist, Consultant

Edge Systems - The HydraFacial Company

Lead the data science and cloud infrastructure team through research and development initiatives related to IoT medical device informatics. Responsible for innovation with product marketing and engineering to improve and enhance technology products and processes. Implemented computer vision capabilities on the HydraFacial skin analyzer device for detection and classification of dermatologic conditions. (NASDAQ IPO: SKIN, now called Beauty Health Co.)

July 2019 - March 2020 Director of Data Science, Disruptive Labs

Ember.AI

Developed a custom artificial intelligence engine for building state-of-the-art natural language processing models for multilingual fraud and compliance-related use cases. Built a large-scale, massively parallel processing architecture using Amazon Web Services and Databricks (Apache Spark) for interactive model training. (IP Acquired by Barrick Gold Corp.)

August 2018 - March 2020 Chief Data Scientist

Cencora *(formerly AmerisourceBergen)*

Supported clinical trial operations for multiple late-stage drug programs. Developed machine learning experiments and data analysis workflows to aid in pharmaceutical client analytics. Created analysis pipelines for the discovery and understanding into patient drug adherence and rare disease drug access. Consulted in company data architecture including master data management and governance for future business strategy to grow value for client accounts.

February 2016 - February 2017 Senior Data Science Lead, Lash Group

Mariner *(a Microsoft Partner)*

Consulted in the development of machine learning experiments, including parametric and non-parametric models, statistical predictions, and data mining. Built Azure cloud-based solutions for data collection, processing, and storage using Microsoft Azure Technologies such as Data Science Virtual Machines, Machine Learning Studio, Azure SQL Database, and more. Designed and created interactive visualizations for both dashboarding and reporting using Microsoft Power BI, Tableau Desktop, & Tableau Server.

October 2014 - February 2016 Data Scientist, Consultant

Caldwell Hospice and Palliative Care

Pioneered the transition from handwritten, paper medical forms to electronic data and served as a database administrator. Worked closely with HIPAA and Medicare/Medicaid guidelines for data compliance.

August 2008 - December 2011 Medical Records Database Administrator

Academic Positions

The University of North Carolina at Charlotte

Performing research across a broad range of fields in the bioinformatics and computational biology space including infectious diseases, epistasis, human genomics, phylogenetics, and AI in drug discovery. Redesigned and taught *Cloud Computing for Data Analysis* (DSBA 6190) for the School of Data Science Master's program. Received a funding grant from Microsoft to modernize and rebuild the course content on the Azure cloud platform.

September 2023 - Present	Visiting Scholar, AI	CIPHER Research Center
January 2019 - December 2025	Associate Faculty, Cloud	School of Data Science
July 2021 - October 2024	University Business Partner	Research and Economic Development
January 2021 - May 2021	Public Health Researcher	Dept. of Public Health
January 2019 - July 2021	Bioinformatics Researcher	Dept. of Bioinformatics and Genomics
April 2018 - December 2018	Postdoctoral Researcher	Dept. of Bioinformatics and Genomics
May 2016 - April 2018	Research Assistant	Dept. of Bioinformatics and Genomics

Northeastern University

Taught a summer session of machine learning to graduate students in the LEVEL analytics program.

May 2015 - August 2015	Guest Lecturer	D'Amore-McKim School of Business

North Carolina New Schools

Responsible for entire technology inventory: ordering, maintenance, management, etc. and maintained school website. Liaison between college (CCC&TI) and high school technology departments. Taught NCVPS mathematics courses, held additional teaching sessions in math and science and SAT & ACT preparation.

April 2014 - October 2014	Technology Facilitator	Caldwell Early College
April 2014 - October 2014	Assistant Mathematics Instructor	Caldwell Early College

Publications

Peer-Reviewed Journal Articles

J29. Sayal Guirales-Medrano, Kary Ocaña, Khaled Obeid, Rachel Alexander, **Colby T. Ford**, and Daniel Janies. Computational Structural Analysis Predicts Host Range Promiscuity and Antiviral Resistance in North American H5N1 Lineages. *Computational and Structural Biotechnology Journal*, April 2026

J28. Nicholas F. Santolla and **Colby T. Ford**. AI-Based Antibody Design Targeting Recent H5N1 Avian Influenza Strains. *Computational and Structural Biotechnology Journal*, June 2025

J27. **Colby T. Ford**, Shirish Yasa, Khaled Obeid, Rafael Jaimes III, Phillip J. Tomezsko, Sayal Guirales-Medrano, Richard Allen White III, and Daniel Janies. Large-Scale Computational Modelling of H5 Influenza Variants Against HA1-Neutralising Antibodies. *Lancet eBioMedicine*, 114, April 2025

J26. Arianna Comendul, Frederique Ruf-Zamojski, **Colby T. Ford**, Pankaj Agarwal, Elena Zaslavsky, German Nudelman, Manoj Hariharan, Aliza Rubenstein, Hanna Pincas, Venugopalan D. Nair, Adam M. Michaleas, Philip D. Fremont-Smith, Darrell O. Ricke, Stuart C. Sealfon, Christopher W. Woods, Kajal T. Claypool, and Rafael Jaimes. Comprehensive guide for epigenetics and transcriptomics data quality control. *STAR Protocols*, 6(1):103607, 2025

J25. **Colby T. Ford**, Jake A. Galler, Yingnan He, Cathrine Young, Beata Gabriela K. Simpson, Chao-Yi Wu, Jake Pfaffenroth, Eh So Wah, Steven E. Arnold, Hiroko H. Dodge, and Sudeshna Das. Using Apple Watches to Monitor Health and Behaviors of Individuals with Cognitive Impairment: A Case Series Study. *The Journals of Gerontology: Series A*, 10 2024

J24. **Colby T. Ford**. PD-1 Targeted Antibody Discovery Using AI Protein Diffusion. *Technology in Cancer Research & Treatment*, 23, 2024

J23. Shirish Yasa, Sayal Guirales-Medrano, Denis Jacob Machado, **Colby T. Ford**, and Daniel A. Janies. Predicting Antibody and ACE2 Affinity for SARS-CoV-2 BA.2.86 and JN.1 with *In Silico* Protein Modeling and Docking. *Frontiers in Virology*, 4, 2024

J22. **Colby T. Ford**, Phillip J. Tomezsko, Avery E. Meyer, Adam M. Michaleas, and Rafael Jaimes III. Human Cytokine and Coronavirus Nucleocapsid Protein Interactivity Using Large-Scale Virtual Screens. *Frontiers in Bioinformatics*, 4, 2024

J21. Nicholas J. Santistevan, **Colby T. Ford**, Cole S. Gilsdorf, and Yevgenya Grinblat. Behavioral and transcriptomic analyses of *mecp2* function in zebrafish. *American Journal of Medical Genetics Part B: Neuropsychiatric Genetics*, e32981, 2024

J20. Tara N. McCray, Vy Nguyen, Jake S. Heins, Elizabeth Ngyuen, Kristen Stewart, **Colby T. Ford**, Calvin Neace Neace, Priyanka Gupta, and David J. Ortiz. Bronchioalveolar organoids: A preclinical tool to screen toxicity associated with antibody-drug conjugates. *Toxicology and Applied Pharmacology*, 485:116886, 2024

J19. The Critical Assessment of Genome Interpretation Consortium. CAGI, the Critical Assessment of Genome Interpretation, establishes progress and prospects for computational genetic variant interpretation methods. *Genome Biology*, 25(1):53, Feb 2024

J18. Hsin Lee, Kwen-Shen Lee, Chia-Hsin Hsu, Chen-Wei Lee, Ching-En Li, Jia-Kang Wang, Chien-Chia Tseng, Wei-Jen Chen, Ching-Chang Horng, **Colby T. Ford**, Andreas Kroh, Omri Bronstein, Hayate Tanaka, Tatsuo Oji, Jih-Pai Lin, and Daniel Janies. Reply to: Embracing the taxonomic and topological stability of phylogenomics. *Scientific Reports*, 14(1):4094, Feb 2024

J17. Daniel Kepple, **Colby T. Ford**, Jonathan Williams, Beka Abagero, Shaoyu Li, Jean Popovici, Delenasaw Yewhalaw, and Eugenia Lo. Comparative transcriptomics reveal differential gene expression among *Plasmodium vivax* geographical isolates and implications on erythrocyte invasion mechanisms. *PLOS Neglected Tropical Diseases*, 18(1):1–21, 01 2024

J16. Cristian M. Galván-Villa, Francisco A. Solís-Marín, Karen Lopez, Janessa Cobb, Leopoldo Díaz-Pérez, Carlos R. Rezende-Ventura, Nataly Slivak, **Colby T. Ford**, and Daniel A. Janies. Occurrence of the Indo-West Pacific starfish *Luidia magnifica* (Echinodermata: Asteroidea) in the Mexican Pacific and a possible introduction to the Caribbean region. *Marine Biodiversity*, 54(1):1, Dec 2023

J15. Hsin Lee, Kwen-Shen Lee, Chia-Hsin Hsu, Chen-Wei Lee, Ching-En Li, Jia-Kang Wang, Chien-Chia Tseng, Wei-Jen Chen, Ching-Chang Horng, **Colby T. Ford**, Andreas Kroh, Omri Bronstein, Hayate Tanaka, Tatsuo Oji, Jih-Pai Lin, and Daniel Janies. Phylogeny, ancestral ranges and reclassification of sand dollars. *Scientific Reports*, 13(1):10199, Jun 2023

J14. **Colby T. Ford**, Shirish Yasa, Denis Jacob Machado, Richard Allen White III, and Daniel A. Janies. Predicting changes in neutralizing antibody activity for SARS-CoV-2 XBB.1.5 using *in silico* protein modeling. *Frontiers in Virology*, 3, 2023

J13. **Colby T. Ford**, Cheikh Cambel Dieng, Anita Lerch, Dickson Doniou, Kovidh Vegesna, Daniel Janies, Liwang Cui, Linda Amoah, Yaw Afrane, and Eugenia Lo. Genetic variations of *Plasmodium falciparum* circumsporozoite protein and the impact on interactions with human immunoproteins and malaria vaccine efficacy. *Infection, Genetics and Evolution*, 110:105418, 2023

J12. Dorcas Bredu, George K. Ahadzi, Donu Dickson, Sherik-fa Anang, Alexander Asamoah, George Asumah, Nana Prepah, Dorcas Obiri Yeboah, **Colby T. Ford**, Eugenia Lo, Keziah Laurencia Malm, and Linda Eva Amoah. Nationwide surveillance of Pfhrp2 exon 2 diversity in *Plasmodium falciparum* circulating in symptomatic malaria patients living in Ghana. *American Journal of Tropical Medicine and Hygiene.*, 2022

J11. **Colby T. Ford**, Denis Jacob Machado, and Daniel A Janies. Predictions of the SARS-CoV-2 Omicron Variant (B.1.1.529) Spike Protein Receptor-Binding Domain Structure and Neutralizing Antibody Interactions. *Frontiers in Virology*, 2, 2022

J10. Anthony Ford, Daniel Kepple, Jonathan Williams, Gabrielle Kolesar, **Colby T. Ford**, Abnet Abebe, Lemu Golassa, Daniel A. Janies, Delenasaw Yewhalaw, and Eugenia Lo. Gene Polymorphisms Among *Plasmodium vivax* Geographical Isolates and the Potential as New Biomarkers for Gametocyte Detection. *Frontiers in Cellular and Infection Microbiology*, 11, 2022

J9. **Colby T. Ford**, Gezahegn Solomon Alemayehu, Kayla Blackburn, Karen Lopez, Cheikh Cambel Dieng, Lemu Golassa, Eugenia Lo, and Daniel Janies. Modeling *Plasmodium falciparum* Diagnostic Test Sensitivity Using Machine Learning With Histidine-Rich Protein 2 Variants. *Frontiers in Tropical Diseases*, 2:28, 2021

J8. Anthony Ford, Daniel Kepple, Beka Raya Abagero, Jordan Connors, Richard Pearson, Sarah Auburn, Sisay Getachew, **Colby T. Ford**, Karthigayan Gunalan, Louis H. Miller, Daniel A. Janies, Julian C. Rayner, Guiyun Yan, Delenasaw Yewhalaw, and Eugenia Lo. Whole genome sequencing of *Plasmodium vivax* isolates reveals frequent sequence and structural polymorphisms in erythrocyte binding genes. *PLOS Neglected Tropical Diseases*, 14(10):1–27, 10 2020

J7. **Colby T. Ford**, Gabriel Lopez Zenarosa, Kevin B. Smith, David C. Brown, John Williams, and Daniel Janies. Genetic Capitalism and Stabilizing Selection of Antimicrobial Resistance Genotypes in *Escherichia coli*. *Cladistics*, 2020

J6. **Colby T. Ford**, Jia Wen, Daniel Janies, and Xinghua Shi. A Parallelized Strategy for Epistasis Analysis Based on Empirical Bayesian Elastic Net Models. *Bioinformatics*, 03 2020. btaa216

J5. **Colby T. Ford** and Daniel Janies. Ensemble machine learning modeling for the prediction of artemisinin resistance in malaria. *F1000Research*, 9(62), 2020

J4. Adriano de Bernardi Schneider, **Colby T. Ford**, Reilly Hostager, John Williams, Michael Cioce, Ümit V. Catalyürek, Joel O Wertheim, and Daniel Janies. StrainHub: A phylogenetic tool to construct pathogen transmission networks. *Bioinformatics*, 08 2019

J3. Wyatt T. Clark, Laura Kasak, Constantina Bakolitsa, Zhiqiang Hu, Gaia Andreoletti, Giulia Babbi, Yana Bromberg, Rita Casadio, Roland Dunbrack, Lukas Folkman, **Colby T. Ford**, David Jones, Panagiotis Katsonis, Kunal Kundu, Olivier Lichtarge, Pier L. Martelli, Sean D. Mooney, Conor Nodzak, Lipika R. Pal, Predrag Radivojac, Castrense Savojardo, Xinghua Shi, Yaoqi Zhou, Aneeta Uppal, Qifang Xu, Yizhou Yin, Vikas Pejaver, Meng Wang, Liping Wei, John Moult, Guoying Karen Yu, Steven E. Brenner, and Jonathan H. LeBowitz. Assessment of predicted enzymatic activity of α-N-acetylglucosaminidase variants of unknown significance for CAGI 2016. *Human Mutation*, 40(9):1519-1529, 2019

J2. **Colby T. Ford**. An integrated phylogeographic analysis of the Bantu migration. *ProQuest Dissertations and Theses*, page 120, 2018

J1. Daniel Janies, **Colby Ford**, Lambodhar Damodaran, and Zacharey Faigen. Spread of Middle East Respiratory Coronavirus: Genetic versus Epidemiological Data. *Online Journal of Public Health Informatics*, 9(1), 2017

Preprints

P7. Phillip Souza and **Colby T. Ford**. Hoike: A Joint-Embedding Predictive Architecture for Transcriptome Data Generation with Diffusion Models. *bioRxiv*, 2026

P6. Seth Rabinowitz, Prbhuv Nigam, Nicholas Santolla, and **Colby T. Ford**. Kukulu: Diffusion-Based Reconstruction of Antibody CDR Loops using a Structure-Aware Joint Embedding Predictive Architecture. *bioRxiv*, 2026

P5. Daniel Janies, Sayal Guirlaes-Medrano, Ana Clecia Dos Santos Silva, and **Colby T. Ford**. A map of the historical spread of Cyclospora cayetanensis with a focus on the United States of America (USA). *medRxiv*, 2026

P4. Beka Abagero, Franck Dumetz, **Colby T. Ford**, Tirusew Tolosa, Daniel Tesefay, Biniyam Lukas, Tassew Shenkutie, Jean Popovici, Delenesaw Yewhalaw, David Serre, and Eugenia Lo. Stage-aware transcriptomics reveals selective haplotype persistence in short-term *ex vivo* cultured *Plasmodium vivax*. *bioRxiv*, 2026

P3. Nicholas Santolla, Trey Pridgen, Prbhuv Nigam, and **Colby T. Ford**. peleke-1: A Suite of Protein Language Models Fine-Tuned for Targeted Antibody Sequence Generation. *bioRxiv*, 2025

P2. **Colby T. Ford**, Rachel Scott, Denis Jacob Machado, and Daniel Janies. Sequencing Data of North American SARS-CoV-2 Isolates Shows Widespread Complex Variants. *medRxiv*, 2021

P1. **Colby T. Ford**, Aneeta Uppal, Conor M. Nodzak, and Xinghua Shi. Prediction of the effect of naturally occurring missense mutations on cellular N-acetyl-glucosaminidase enzymatic activity. *bioRxiv*, 2019

Conference Papers and Presentations

C15. Daniel Janies and **Colby T. Ford**. Accelerating Biodefense: AI-Driven Structural Biology for Rapid Development of Medical Countermeasures. *Military Health System Research Symposium*, August 2025. MHSRS-25-13625

C14. **Colby T. Ford**. Model. Package. Deploy. Repeat: A DevOps Approach to Biomolecular Scalability. *Boston Protein Design and Modeling Club Lecture*, July 2025

C13. **Colby T. Ford** and Shirish Yasa. Large-Scale AI and Physics Modeling of H5N1 Influenza and Neutralizing Antibody Interactions. *ASM Microbe*, June 2025

C12. Daniel Janies, Shirish Yasa, **Colby T. Ford**, Jannatul Ferdous, William Taylor, April Harris, Sam Kunkleman, Juan Bolanos, Kevin Lambirth, Denis Jacob Machado, Cynthia Gibas, and Jessica Schlueter. Evaluation of SARS-CoV-2 response at the University of North Carolina (UNC) at Charlotte using percent positivity data and viral genomic sequence data. *BioCARLA 2024*, October 2024

C12. **Colby T. Ford** and Daniel Janies. Rapid and Flexible Development of Medical Countermeasures with Computational Structural Biology and Artificial Intelligence. *DTRA Chemical and Biological Defense Science & Technology Conference*, December 2024

C11. **Colby T. Ford**. Lessons in Scalability: Cloud and HPC-Driven Computational Biology. *MITLL Biotechnology & Human Systems Seminar*, 2024

C10. **Colby T. Ford**, Eh So Wah, Jake Pfaffenroth, and Fidel Henriquez. Scaling Digital Biomarker Discovery with the Cloud and Wearable Devices. *Alzheimer's & Dementia*, 19(S18):e078069, 2023

C9. Kira Chiles, Candace S. Brown, and **Colby T. Ford**. The effects of exercise during a pandemic. *UNC Charlotte Undergraduate Research Conference*, 2023

C8. Adriano de Bernardi Schneider, **Colby T. Ford**, Jakob McBroome1, Jennifer Martin, Daniel Janies, Yatish Turakhia, and Russel Corbett-Detig. Understanding the spread of SARS-CoV-2 clusters through an integrated pipeline using UShER, Cluster Tracker and StrainHub. *Genetics Society of America - Population, Evolutionary, and Quantitative Genetics Conference*, 2022

C7. Cheikh Cambel Dieng, **Colby T. Ford**, Dickson Doniou, Jennifer Huynh, Anita Lerch, Daniel Janies, Linda Amoah, Yaw Afrane, and Eugenia Lo. Population structure and selection-mediated changes in *Plasmodium falciparum* by next-generation sequencing. *Annual Meeting of the American Society of Tropical Medicine and Hygiene*, 2020

C6. **Colby T. Ford**. Visualizing transmission networks of pathogens using phylogenetic data with StrainHub. *I Meeting of Systematics, Biogeography, and Evolution - Symposium of Virology in the SARS-CoV-2 era*, 2020

C5. Daniel Janies, **Colby T. Ford**, Kevin Smith, Gabriel Lopez Zenarosa, and John Williams. Evolution of gain and loss of antimicrobial resistance genes in *Escherichia coli*. *XXXVIII Annual Meeting of the Willi Hennig Society*, 2019

C4. Kevin Smith, **Colby T. Ford**, Gabriel Lopez Zenarosa, John Williams, and Daniel Janies. Phylogenetic analysis of the genetic variation of multi-drug resistant *Escherichia coli*. *National Council on Undergraduate Research*, 2019

C3. Jia Wen, **Colby T. Ford**, Daniel Janies, and Xinghua Shi. New strategies toward scaling up epistasis analysis on large-scale genomic datasets. *ACM Conference on Bioinformatics, Computational Biology, and Health Informatics*, 2018

C2. **Colby T. Ford** and Andy Lathrop. Predictive modeling of vegetation density using R and a cloud data platform. *Analytics for Social Good, University of Cincinnati*, 2017

C1. **Colby T. Ford**, Ming Xue, Peter M. Whiteley, Ward Wheeler, Daniel A. Janies, and Xinghua Shi. Visualizing linguistic disparity of Uto-Aztecan languages and Bantu languages. *Society for Anthropological Sciences Annual Meeting*, 2016

Software and Coding

S5. **Colby T. Ford**, Samee Ullah, Dinler Amaral Antunes, and Tarsis Gesteira Ferreira. PyMOLfold: Interactive Protein and Ligand Structure Prediction in PyMOL, 2025

S4. **Colby T. Ford**. *ahab-lib* - Python library and CLI for interacting with the *ahab* cloud API and Kubernetes system, April 2024

S3. **Colby T. Ford**. *msgen* - R functions for interfacing with the Microsoft Genomics service in Azure, January 2021

S2. Adriano de Bernardi Schneider, **Colby T. Ford**, Reilly Hostager, John Williams, Michael Cioce, Ümit V. Catalyürek, Joel O Wertheim, and Daniel Janies. StrainHub: A phylogenetic tool to construct pathogen transmission networks. *Bioinformatics*, 08 2019

S1. **Colby T. Ford**, Jia Wen, Daniel Janies, and Xinghua Shi. A Parallelized Strategy for Epistasis Analysis Based on Empirical Bayesian Elastic Net Models. *Bioinformatics*, 03 2020. btaa216

Books, eBooks, and Chapters

C2. **Colby T. Ford**. Generative AI for Antibody Discovery and Engineering: Emerging Models, Design Strategies, and Future Directions. In *Advanced Antibody Engineering - Monoclonal and Polyclonal Technologies in Therapeutics and Diagnostics*. IntechOpen, London, 2026. ISBNs: 978-1-80632-798-0, 978-1-80632-799-7, 978-1-80632-800-0, DOI: 10.5772/intechopen.96530. *In progress.*

B3. **Colby T. Ford**. *Building Agentic Solutions with Microsoft Foundry*. O'Reilly Media, September 2026. ISBN: 979-8-341-67332-8

B2. **Colby T. Ford**. *Genomics in the Azure Cloud*. O'Reilly Media, Dec 2022. ISBN: 978-1-098-13904-9

E2. **Colby T. Ford** and Larry Baker. *Building a Genomics Data Lake in Azure*. BlueGranite, Jan 2021. URL: https://zenodo.org/record/4474520, DOI: 10.5281/zenodo.4474520

C1. Cheikh Cambel Dieng, **Colby T. Ford**, Jennifer Huynh, Linda E. Amoah, Yaw A. Afrane, Daniel A. Janies, and Eugenia Lo. Progress in *Plasmodium* genomics and current challenges in malaria control. In *Current Topics and Emerging Issues in Malaria Elimination*. IntechOpen, London, 2021. ISBNs: 978-1-83968-484-5, 978-1-83968-483-8, DOI: 10.5772/intechopen.96530

E1. **Colby T. Ford**. *Sparkitecture - A collection of "cookbook-style" scripts for simplifying data engineering and machine learning in Apache Spark.* GitBook, October 2019

B1. **Colby Ford**. *Caesura - late-intermediate piano technique book.* Lulu, March 2010. ISBN: 978-0-557-36832-7

Technical Articles

TA61. **Colby T. Ford**. Evaluating a Ternary Protein Language Model Architecture for Rapid Antibody Sequence Generation, August 2026. Medium

TA60. **Colby T. Ford**. Building Agentic Experiences with Website Search Capabilities in Microsoft Foundry, July 2026. Medium

TA59. **Colby T. Ford**. Connecting Azure Kubernetes Clusters to Tailscale Networks, June 2026. Medium

TA58. **Colby T. Ford**. Building Multi-agentic Bioworkflows with Microsoft Foundry and Nextflow, May 2026. Medium

TA57. **Colby T. Ford**. Connecting OpenClaw to Models in Microsoft Foundry, April 2026. Medium

TA56. **Colby T. Ford**. Structuring Proteomics Data from Multiple Lab Vendors in Cloud Data Warehouses, March 2026. Medium

TA55. **Colby T. Ford**. Deploying Custom Microsoft Foundry Agents as Shiny Python Apps in Azure, February 2026. Medium

TA54. **Colby T. Ford**. Building Data-Heavy Bioinformatics APIs using Serverless Azure Functions and Cosmos DB, January 2026. Medium

TA53. **Colby T. Ford**. Challenges in Effective Candidate Selection in AI-Based Antibody Design - Our Nipah Virus Experience, December 2025. Medium

TA53. **Colby T. Ford**. Foundry: The Top AI Announcement from Microsoft Ignite 2025, November 2025. Medium

TA52. **Colby T. Ford**. Run Azure AI Models on your Laptop with Foundry Local and Olive, October 2025. Medium

TA51. **Colby T. Ford**. Getting Started with Agentic AI with Azure Foundry Agent Service on Microsoft Fabric Data, September 2025. Medium

TA50. **Colby T. Ford**. 5 Tips for Fine-Tuning Protein Language Models (PLMs) on Azure Machine Learning, August 2025. Medium

TA49. **Colby T. Ford**. Private Protein Folding with Multiple Sequence Alignments in Azure AI Foundry, July 2025. Medium

TA48. **Colby T. Ford**. Creating an MCP Server for AI-Powered Bioinformatics Data Queries, June 2025. Medium

TA47. **Colby T. Ford**. Automating Genomics Data Management with Azure Storage Actions, May 2025. Medium

TA46. **Colby T. Ford**. Reading Genomics Data in Snowflake from Azure Storage Accounts (such as TCGA Expression Data), April 2025. Medium

TA45. **Colby T. Ford**. 3 Tips for Building Docker Images of AI Models or Complex Scientific Tools, March 2025. Medium

TA44. **Colby T. Ford**. New Week, New Molecular AI Models: BioEmu-1 and Evo 2, February 2025. Medium

TA43. **Colby T. Ford**. 4 Ways to Run Your Own DeepSeek AI Instance Today, January 2025. Medium

TA42. **Colby T. Ford**. Exploring the Azure Cloud's Free Service Offerings, January 2025. Medium

TA41. **Colby T. Ford**. Running AlphaFold3 At-Scale on High Performance Computing Clusters, December 2024. Medium

TA40. **Colby T. Ford**. Microsoft Ignite 2024: Top Announcements for Life Science Workloads, November 2024. Medium

TA39. **Colby T. Ford**. Quick Note: Health and Life Science Models are now in Azure AI Studio, October 2024. Medium

TA38. **Colby T. Ford**. Easily Scaling Container Workloads using Automatic Clusters in Azure Kubernetes Service, September 2024. Medium

TA37. **Colby T. Ford**. Building a Custom AI Chatbot with Azure AI Studio, August 2024. Medium

TA36. **Colby T. Ford**. Deploy Protein Language Models using Ollama, July 2024. Medium

TA35. **Colby T. Ford**. Building GPU-Enabled Clusters in Azure Kubernetes Service, June 2024. Medium

TA34. **Colby T. Ford**. Microsoft Build 2024: 4 Hot New AI Announcements!, May 2024. Medium

TA33. **Colby T. Ford**. Tips for Succeeding in the Microsoft for Startups Founders Hub Program, April 2024. Medium

TA32. **Colby T. Ford**. Simple Syncing Between Azure Data Lake and Synology NAS, February 2024. Medium

TA31. **Colby T. Ford**. Transfer Sequencing Data from Lab Vendors using sFTP and Azure Synapse Pipelines, January 2024. Medium

TA30. **Colby T. Ford**. 2023: A Biotech Year-End Review, December 2023. Medium

TA29. **Colby T. Ford**. 5 Exciting Azure Announcements from Microsoft Ignite 2023, November 2023. Medium

TA28. **Colby T. Ford**. Exploring AI-Based Protein Sequence Generation with EvoDiff from Microsoft Research, October 2023. Medium

TA27. **Colby T. Ford**. GPT on my Genome: Building a QnA System for Genomic Variants with Azure OpenAI, September 2023. Medium

TA26. **Colby T. Ford**. 4 Tips for Qualifying your Azure Data Lake for GxP use in the Life Sciences, August 2023. Medium

TA25. **Colby T. Ford**. Predicting the molecular interactions of lecanemab, the new FDA-approved monoclonal antibody for Alzheimer's disease., July 2023. Medium

TA24. **Colby T. Ford**. The Future of Manufacturing is Bio: Things I Learned in the Millyard., June 2023. Medium

TA23. **Colby T. Ford**. How Azure OneLake will revolutionize the way we manage and use -omics data, May 2023. Medium

TA23. **Colby T. Ford**. Deploying Databricks Dolly as an API on Azure Functions, April 2023. Medium

TA22. **Colby T. Ford**. Deploying Azure OpenAI and Building a Custom Science Article Recommender App with ChatGPT, March 2023. Medium

TA21. **Colby T. Ford**. What does ChatGPT think about using the Azure cloud for genomics?, January 2023. Medium

TA20. **Colby T. Ford**. 5 Quick Things You Didn't Know About the Microsoft Cloud for Healthcare, November 2022. Medium

TA19. **Colby T. Ford**. How to Deploy Shiny Apps in Azure and use your Domain Name, October 2022. Medium

TA18. **Colby T. Ford**. Using Microsoft Dev Box and Azure Compute Galleries for Bioinformatics, August 2022. Medium

TA17. **Colby T. Ford**. A Deep Dive Look Into Global Monkeypox Transmissions - from January to July 2022., July 2022. Medium

TA16. **Colby T. Ford**. Deploying Azure Databricks with a Virtual Network and Network Security Group, July 2022. Medium

TA15. **Colby T. Ford**. Quick Note: What's up with the new SARS-CoV-2 BA.2.75 and BA.5 variants?, July 2022. Medium

TA14. **Colby T. Ford**. Am I Hallucinating or Can AI Now Design Cancer-Curing Antibodies?, June 2022. Medium

TA13. **Colby T. Ford**. Scaling Genomics in the Cloud with Microsoft Azure, May 2022. Medium

TA12. **Colby T. Ford**. How to predict many protein structures with AlphaFold2 at-scale in Azure Machine Learning, January 2022. Medium

TA11. **Colby T. Ford**. Predicted Protein Interactions of the SARS-CoV-2 B.1.1.529 Variant with Neutralizing Antibodies, November 2021. Medium

TA10. **Colby T. Ford**. Protein Structure Prediction of the new B.1.1.529 SARS-CoV-2 Spike Variant with AlphaFold2, November 2021. Medium

TA9. **Colby T. Ford**. How Do We Fix Peer Review? (Because It's Broken...), October 2021. Medium

TA8. **Colby T. Ford**. I Asked GPT-3 to Write Some Chocolate Chip Cookie Recipes. . .and I Baked Them), June 2021. Medium

TA7. **Colby T. Ford**. What is a "variant" as it relates to COVID-19? (for non-geneticists), February 2021. Medium

TA6. **Colby T. Ford**. Automated Alignment and Variant Calling in Azure using the Microsoft Genomics service and the *msgen* R package, January 2021. Medium

TA5. **Colby T. Ford**. Better Evaluate COVID-19 Test Performance using Bayes Factors, December 2020. Medium

TA4. **Colby T. Ford**. Assessment of retail out-of-stock conditions using statistical inference. Technical report, Mariner, 2016

TA3. **Colby T. Ford** and Wayne Snyder. Revenue protection using machine learning for utilities management. Technical report, Mariner, 2015

TA2. **Colby T. Ford**. The allure of machine learning, now within reach in Microsoft Azure. Technical report, Mariner, 2015

TA1. **Colby T. Ford**. Demand forecasting using machine learning to reduce working capital. Technical report, Mariner, 2015

Professional Blog Posts and Case Studies

B31. **Colby T. Ford**. Supporting the Fight Against Cancer, Tuple Unveils Anti-PD-1 Antibodies Fully Designed by AI, 2024. *Tuple Case Study*

B31. **Colby T. Ford**. MIT Lincoln Lab and Tuple Achieve Immense Scalability in SARS-CoV-2 Molecular Modeling with High-Performance Computing, 2023. *Tuple Case Study*

B30. **Colby T. Ford**. Case Study: Scaling Rare Disease Research in the Azure Cloud, 2022. *Tuple Case Study*

B29. **Colby T. Ford**. Using Azure CycleCloud and Batch for Scalable HPC Workloads, 2021. *BlueGranite Technical Blog*

B28. **Colby T. Ford**. Microsoft AI: Look, Listen, Innovate! Hackfest - BlueGranite's Smart City Solution, 2021. *BlueGranite Technical Blog*

B27. **Colby T. Ford**. Query Millions of Genomic Variants in Minutes with Azure Synapse, 2021. *BlueGranite Technical Blog*

B26. **Colby T. Ford**. Review: How 3 Life Science Organizations Harness the Power of the Cloud, 2021. *BlueGranite Technical Blog*

B25. **Colby T. Ford**. Introducing BlueGranite's Azure Data Factory Connector for Illumina BaseSpace, 2021. *BlueGranite Technical Blog*

B24. **Colby T. Ford**. Event Recap: Shape your Future with Azure Data and Analytics, 2020. *BlueGranite Technical Blog*

B23. **Colby T. Ford**. Reading Bioinformatics and Genomics Files in Power BI, 2020. *BlueGranite Technical Blog*

B22. Jon Gore, Thomas J. Weinandy, and **Colby T. Ford**. Take Back Your Data to Gain a Competitive Advantage, 2020. *BlueGranite Technical Blog*

B21. **Colby T. Ford**. A BlueGranite Blog Post Written (Mostly) by AI, 2020. *BlueGranite Technical Blog*

B20. **Colby T. Ford**. Give Your Genomics Pipeline a *Glow* Up in Azure Databricks, 2020. *BlueGranite Technical Blog*

B19. **Colby T. Ford**. SIR Modeling on Azure: COVID-19 Hospital Impact Model for Epidemics, 2020. *BlueGranite Technical Blog*

B18. **Colby T. Ford**. Let's Talk About COVID-19, 2020. *BlueGranite Technical Blog*

B17. **Colby T. Ford**. Comparing Azure Machine Learning Service and Azure Databricks, 2020. *BlueGranite Technical Blog*

B16. **Colby T. Ford**. Recap of rstudio::conf(2020) for Data Science and Machine Learning, 2020. *BlueGranite Technical Blog*

B15. **Colby T. Ford**. Scaling your Genomics Pipeline in the Cloud with Azure Databricks, 2019. *BlueGranite Technical Blog*

B14. **Colby T. Ford**. Migrating & Scaling Machine Learning Models to Azure Databricks for Cloud-Powered AI, 2019. *BlueGranite Technical Blog*

B13. **Colby T. Ford**. Introducing the Databricks Unified Analytics Platform for Genomics, 2018. *BlueGranite Technical Blog*

B12. **Colby T. Ford**. Recap: Spark+AI Summit 2018, 2018. *BlueGranite Technical Blog*

B11. **Colby T. Ford**. Cognitive Services Showcase: API Search Tools, 2018. *BlueGranite Technical Blog*

B10. **Colby T. Ford**. Let Azure do the Heavy Lifting on Your AI Workload, 2018. *BlueGranite Technical Blog*

B9. **Colby T. Ford**. Recap of rstudio::conf 2018, 2018. *BlueGranite Technical Blog*

B8. **Colby T. Ford**. Microsoft Azure & Databricks = Cloud-Scale Spark Power, 2017. *BlueGranite Technical Blog*

B7. **Colby T. Ford**. Maximize Your Customer Retention by Predicting Customer Churn, 2017. *BlueGranite Technical Blog*

B6. **Colby T. Ford**. Become the Maestro of your Genomics Workflow with Bioconductor and Microsoft R Server, 2017. *BlueGranite Technical Blog*

B5. **Colby T. Ford**. Publishing Predictive Web Services with Microsoft R Server, 2017. *BlueGranite Technical Blog*

B4. **Colby T. Ford**. Data Visualization for Bioinformatics with R in Power BI, 2017. *BlueGranite Technical Blog*

B3. **Colby T. Ford**. Webinar Recap: Distributed Computing & R Server, 2017. *BlueGranite Technical Blog*

B2. **Colby T. Ford**. SAS Enterprise Guide vs. Microsoft Azure Machine Learning, 2017. *BlueGranite Technical Blog*

B1. **Colby T. Ford**. ImpoRting and ExpoRting: Getting Data Into and Out of R, 2017. *BlueGranite Technical Blog*

Conferences, Training, Press, and Speaking Engagements

Nov 2026	Microsoft Ignite - Foundry	*Speaker*	San Francisco, CA
Sep 2026	AI in GxP East Conference - LLM Adoption in Pharma	*Speaker*	Raleigh, NC
Sep 2026	Charlotte Biomedical Sciences Symposium	*Speaker*	Charlotte, NC
Jun 2026	Microsoft Build - Cloud Platform & Data	*Speaker*	San Francisco, CA
May 2026	JCSU DataSmiths Bootcamp - AI in Computational Drug Design	*Instructor*	Charlotte, NC
May 2026	AI//Forward Research & Innovation Symposium	*Speaker*	Charlotte, NC
May 2026	AI in GxP West Conference - Governable AI in Pharma	*Speaker*	San Diego, CA
Apr 2026	Nextflow Summit 2026	*Speaker*	Boston, MA
Jan 2026	Microsoft AI Tour - Fabric, Databricks, & Foundry Workshop	*Speaker*	New York, NY
Nov 2025	Microsoft Ignite - AI Infrastructure and HPC	*Speaker*	San Francisco, CA
Nov 2025	Evolved AIxBio Global Sprint Hackathon	*Participant*	Online
Oct 2025	7th Molecular Machine Learning Conference (MoML) @ MIT	*Presenter*	Cambridge, MA
Sep 2025	posit::conf(2025)	*Attendee*	Atlanta, GA
Sep 2025	Charlotte Biomedical Sciences Symposium	*Speaker*	Charlotte, NC
Jul 2025	WBTV Interview on AI-based Avian Flu Research	*Interviewee*	Charlotte, NC
Jul 2025	Boston Protein Design and Modeling Club Lecture @ Harvard	*Speaker*	Cambridge, MA
Jun 2025	ASM Microbe 2025	*Speaker*	Los Angeles, CA
May 2025	Microsoft Build - Open-Source/Linux on Azure	*Speaker*	Seattle, WA
Apr 2025	US News Report on H5N1 Avian Influenza	*Interviewee*	Online
Mar 2025	Techstars - Startup Weekend AI & Life Sciences	*Speaker*	Wins-Salem, NC
Jan 2025	Microsoft AI Tour - Azure Application Platform	*Speaker*	New York, NY
Jan 2025	Microsoft Empire State of AI Event	*Attendee*	New York, NY
Dec 2024	DTRA Chemical and Biological Defense Science & Tech Conf.	*Presenter*	Ft. Lauder., FL
Nov 2024	Microsoft Ignite	*Host*	Chicago, IL
Nov 2024	UNCC Center for Humane AI Studies Seminar	*Speaker*	Charlotte, NC
Oct 2024	UNCC DSBA 6211 Guest Lecture	*Speaker*	Charlotte, NC
Oct 2024	8th Int. Conf. on Drug Discovery	*Speaker*	Boston, MA
Oct 2024	MITLL Biotechnology & Human Systems Seminar	*Speaker*	Boston, MA
Sep 2024	UNCC Biomedical Sciences Symposium	*Speaker*	Charlotte, NC
Aug 2024	posit::conf(2024)	*Attendee*	Seattle, WA
Jul 2024	AAIC 2024 Panel on LLMs in Dementia Care	*Host/Speaker*	Philadelphia, PA
May 2024	Nextflow Summit 2024	*Speaker*	Boston, MA
May 2024	Microsoft Build	*Attendee*	Online
May 2024	NIH Clinical and Translational Serology Task Force Meeting	*Speaker*	Online
May 2024	London Biotechnology Show	*Speaker*	London, UK
Mar 2024	Leaders of B2B Podcast - Episode 212	*Interviewee*	Online
Mar 2024	Microsoft MVP Global Summit	*Awardee*	Redmond, WA
Mar 2024	Alz. Assoc. AI in Dementia Care Webinar	*Panelist*	Online
Feb 2024	UNCC DSBA 3000 Lecture on Git	*Instructor*	Charlotte, NC
Jan 2024	Microsoft AI Tour - Azure Machine Learning Studio	*Speaker*	New York, NY
Nov 2023	Microsoft Ignite	*Attendee*	Redmond, WA
Oct 2023	UNCC Seminar: Gitting Started with Git	*Instructor*	Charlotte, NC
Sep 2023	posit::conf(2023)	*Speaker*	Chicago, IL
Jul 2023	Alzheimer's Association International Conference 2023	*Presenter*	Amsterdam, NL
Jun 2023	ARMI National Conference on the Future of Biofabrication	*Speaker*	Manchester, NH
Jun 2023	MITLL Biotechnology and Resilient Human Systems Workshop	*Attendee*	Boston, MA
Jun 2023	Data + AI Summit 2023	*Attendee*	San Francisco, CA
Apr 2023	Microsoft MVP Global Summit	*Awardee*	Redmond, WA
Jan 2023	NC HIMSS Event	*Participant*	Charlotte, NC

Nov 2022	Microsoft Cloud for Healthcare Event (TTT)	*Participant*	Seattle, WA
Oct 2022	UNCC Seminar: Gitting Started with Git	*Instructor*	Charlotte, NC
Jul 2022	Alzheimer's Association International Conference 2022	*Panelist*	San Diego, CA
Jul 2022	rstudio::conf(2022)	*Attendee*	Washington DC
Jul 2022	Microsoft Inspire	*Attendee*	Online
Feb 2022	United Nations ITU AI for Good Programme	*Speaker*	Online
Jan 2022	WIRED Interview on Omicron Modeling	*Interviewee*	Online
Dec. 2021	The Economist Interview on Omicron Modeling	*Interviewee*	Online
Nov 2021	Microsoft Ignite	*Attendee*	Online
Oct 2021	Microsoft Research Summit	*Attendee*	Online
Jul 2021	Alzheimer's Association International Conference 2021	*Attendee*	Denver, CO
Jul 2021	Microsoft Ignite 2021	*Attendee*	Online
May 2021	Data + AI Summit 2021	*Attendee*	Online
Mar 2021	Microsoft AI: Look, Listen, Innovate! Solution Hackathon	*Participant*	Online
Jan 2021	Lenoir NewsTopic Interview	*Interviewee*	Online
Jan 2021	rstudio::global(2021)	*Attendee*	Online
Jan 2021	BlueGranite AI-in-a-Day Training Event	*Speaker*	Online
Oct 2020	BlueGranite Loan Risk Analysis Webinar	*Speaker*	Online
Sep 2020	TEDxUNCCharlotte	*Speaker*	Charlotte, NC
Aug 2020	BlueGranite Scaling Genomic Analyses in Azure Webinar	*Speaker*	Online
Jul 2020	SBE Symposium: Virology in the SARS-CoV-2 era	*Speaker*	Online
Jun 2020	Spark + AI Summit 2020	*Speaker/Panelist*	Online
May 2020	UCSD HIV Dynamics Conference	*Speaker*	Online
May 2020	CBS News COVID-19 Interview	*Interviewee*	Online
Apr 2020	WBTV COVID-19 Interview	*Interviewee*	Online
Jan 2020	rstudio::conf(2020)	*Attendee*	San Francisco, CA
Jan 2020	PyData Charlotte	*Speaker*	Charlotte, NC
Nov 2019	UNCC Data Science Initiative Git Workshop	*Speaker*	Charlotte, NC
Oct 2019	UNCC CCI Ph.D. Open House Panel	*Panelist*	Charlotte, NC
Jul 2019	Databricks Retail Webinar	*Speaker*	Online
Jun 2019	Azure Databricks Training Event	*Speaker*	Detroit, MI
Jun 2019	Azure Databricks Training Event	*Speaker*	Charlotte, NC
Jun 2019	BlueGranite Azure Databricks Retail Webinar	*Speaker*	Online
May 2019	CECHS Commencement	*Speaker*	Lenoir, NC
Apr 2019	Azure Databricks Training Event	*Speaker*	Chicago, IL
Mar 2019	Microsoft Azure AI Hackfest	*Attendee*	New York, NY
Jan 2019	rstudio::conf(2019)	*Attendee*	Austin, TX
Sep 2018	Microsoft Ignite	*Attendee*	Orlando, FL
Sep 2018	Big Data Ignite	*Speaker*	Grand Rapids, MI
Jun 2018	Spark + AI Summit 2018	*Attendee*	San Francisco, CA
Jan 2018	rstudio::conf(2018)	*Attendee*	San Diego, CA
Nov 2017	Nissan Analytics Expo	*Speaker*	Nashville, TN
Oct 2017	BlueGranite Retail Webinar	*Speaker*	Online
Mar 2017	BlueGranite Distributed Computing Webinar	*Speaker*	Online
Mar 2017	Society for Applied Anthropology Conference	*Speaker*	Santa Fe, NM
Feb 2017	Analytics for Social Good - U. of Cincinnati	*Speaker*	Cincinnati, OH
Sep 2016	Advanced Pharma Analytics	*Attendee*	Newark, NJ
Sep 2015	Microsoft Cortana Analytics Conference	*Attendee*	Seattle, WA
Nov 2014	Microsoft Roadmap Event	*Speaker*	Charlotte, NC

Skills

Languages:	Python • R • SQL • Javascript • SAS • Visual Basic
High Performance Computing:	Spark (Databricks) • Kubernetes • CUDA • Slurm • MPI
Cloud Computing:	Microsoft Azure • AWS • Docker • Terraform • Bicep/ARM
Artificial Intelligence:	Transformers • Diffusion • Microsoft Foundry • Agents • PEFT/LoRA
Visualization:	Tableau • Power BI • Shiny • ggplot2 • D3.js
Markup/Web:	Markdown • $\LaTeX$ • Quarto • HTML5 • CSS3
Bioinformatics:	RNA-seq • scRNA-seq • WES/WGS • Phylogenetics • Haplotyping
Computational Biology:	Antibody Design • Protein Docking • Drug Sensitivity Modeling

Professional Memberships

Societies

2013-Present	The Society for Industrial and Applied Mathematics
2014-Present	The American Statistical Association
2016-2017	The Society for Anthropological Sciences
2018-2019	American Association for the Advancement of Science
2020-Present	International Society for Computational Biology
2021-Present	International Society for Infectious Diseases
2021-2025	International Society to Advance Alzheimer's Research and Treatment
2025-Present	American Society for Microbiology

Boards

2015-2017	UNCC Data Science Initiative Advisory Board
2015-2016	Northeastern University LEVEL Advisory Board
2022-2023	American Cancer Society - Associate Board of Advisors
2025-Present	Piper Strong Pediatric Cancer Foundation - Board of Advisors
2026-Present	BioCytics - Scientific Advisory Board

Student Advising and Mentoring

Jan 2019 - May 2019	Jainmary Jose	Cloud Computing Teaching Assistant	UNCC
Aug 2019 - Dec. 2019	Anjali Khushalani	Cloud Computing Teaching Assistant	UNCC
Jan 2020 - May 2020	Kovidh Vegesna	Data Science Practicum Intern	UNCC
May 2020 - Sep. 2020	Somesh Kale	Developer Intern	Amissa
May 2020 - Sep. 2020	Heet Detroja	Developer Intern	Amissa
May 2020 - Sep. 2020	Chaitanya Darade	Developer Intern	Amissa
Sep 2020 - May 2021	Adesoji Ademiluyi	Bioinformatics Research Assistant	UNCC
Sep 2020 - Oct. 2021	Eh So Wah	Developer Intern	Amissa
Jan 2022 - May 2022	Paige Oldiges	Cloud Computing Teaching Assistant	UNCC
Jun 2022 - Sep. 2022	Carter DeMordaunt	Bioinformatics Intern	Tuple
Jan 2023 - May 2023	Utwej Sai Nalluri	Cloud Computing Teaching Assistant	UNCC
Aug 2024 - Dec. 2024	Alexander Palmer	Cloud Computing Teaching Assistant	UNCC
Jan 2025 - May 2025	Nicholas Santolla	Data Science Practicum Intern	UNCC
May 2025 - Aug. 2025	Trey Pridgen	Bioinformatics/AI Intern	Tuple
May 2025 - Aug. 2025	Prbhuv Nigam	Bioinformatics/AI Intern	Tuple
Aug 2025 - Dec. 2025	Geetha Marri	Cloud Computing Teaching Assistant	UNCC
May 2026 - Aug. 2026	Seth Rabinowitz	Bioinformatics/AI Intern	Tuple
May 2026 - Aug. 2026	Prbhuv Nigam	Bioinformatics/AI Intern	Tuple
May 2026 - Aug. 2026	Phillip Souza	Bioinformatics/AI Intern	Tuple

Awards

Apr 2022	40 under 40	Charlotte Business Journal
Apr 2022	Impact Award	Seagen, Inc.
Mar 2023	Alumnus of the Year	UNC Charlotte
Apr 2023	Most Valuable Professional (MVP) - Azure	Microsoft
Jun 2023	Impact Award	Seagen, Inc.
Feb 2024	The Fire Award Honoree (Amissa)	CharlotteInno
Jul 2024	Most Valuable Professional (MVP) - Azure + AI	Microsoft
Jul 2024	Nextflow Ambassador	Seqera
Jul 2025	Most Valuable Professional (MVP) - Azure ML, AI, HPC Infra	Microsoft
Jan 2026	Global 100 - Best Biotechnology Business of the Year 2026 (Tuple)	GPMG UK
Jul 2026	Most Valuable Professional (MVP) - Foundry, Azure AI + HPC Infra	Microsoft

Certifications

Technical Certifications

May 2018	Databricks Developer - Apache Spark 2.x for Python	Certification #: 53873
Nov 2019	Microsoft Azure Data Scientist Associate (DP-100)	Exam Writer
Sep 2020	Microsoft Data Science Partnership Program	Custom Core Certification
Oct 2020	Microsoft Azure AI Engineer Associate (AI-100)	Certification #: H532-5207
Oct 2020	Microsoft Azure AI Fundamentals (AI-900)	Certification #: H533-2414
Nov 2020	Databricks Partner Program - Developer Champion	Certification #: 917207
Feb 2021	Microsoft Certified Trainer	Certification #: 17875868
Aug 2025	Snowflake SnowPro Core	Cert #: S107097-250809-COF
Oct 2025	Microsoft Azure ML Engineer Associate (DP-100)	Exam Writer
Jan 2026	Microsoft Azure MLOps Engineer Associate (AI-300)	Exam Writer; Cert #: 05A528-141AJ3
Apr 2026	Microsoft Azure Databricks Data Eng. Assoc. (DP-750)	Exam Writer; Cert #: F4A35P-361D2B
Sep 2026	Microsoft Azure Multi-Agent AI Solutions (AI-500)	Exam Writer

Laboratory/Research Certifications

Nov 2020	CITI Program - Principal Investigators BSS	Record ID: 39539394
Feb 2023	CITI Program - Biomedical Sciences: Responsible Conduct of Research	Record ID: 54594415
Feb 2023	CITI Program - Biomedical Investigators and Research Study Team	Record ID: 54601364
Nov 2024	CITI Program - Social & Behavioral Research (IRB)	Record ID: 65751859
Nov 2020	CITI Program - Institutional Biosafety Committee (IBC) Members BSS	Record ID: 39539396
Jun 2021	CITI Program - Good Clinical Practices (GCP)	Record ID: 43096988
Feb 2023	CITI Program - Biomedical Sciences: Responsible Conduct of Research	Record ID: 54594415
Nov 2020	CITI Program - Shipping and Transport of Regulated Biological Materials	Record ID: 50646437
Apr 2024	CITI Program - Biomedical Research (IRB)	Record ID: 58586217
Dec 2024	CITI Program - Data or Specimens Only Research (IRB)	Record ID: 66353578
Dec 2024	CITI Program - Animal Biosafety	Record ID: 71427713
Oct 2025	CITI Program - Principal Investigators BSS	Record ID: 71427711
Oct 2025	CITI Program - Laboratory Personnel BSS	Record ID: 71427712
Oct 2025	CITI Program - Recombinant and Synthetic Nucleic Acid Molecules	Record ID: 71427714
Oct 2025	CITI Program - USDA Permits	Record ID: 71427715

Peer Review Activity

Scientific Journals

American Chemical Society	ACS Omega
BioMed Central	Genomic Data
Frontiers in Pediatrics	Pediatric Infectious Diseases
Frontiers in Chemistry	Chemical Biology
Frontiers in Immunology	Vaccines and Molecular Therapeutics
Frontiers in Oncology	Cancer Molecular Targets and Therapeutics
Frontiers in Physiology	Red Blood Cell Physiology
Frontiers in Medicine	Infectious Diseases: Pathogenesis and Therapy
Frontiers in Public Health	Infectious Diseases: Epidemiology and Prevention
Nature	Scientific Reports
PLOS	Computational Biology
PNAS	Biophysics and Computational Biology
Science Advances	Biomedicine and Life Sciences
Wiley	Biochemistry Research International

Grant Panels

NIH SBIR/STTR	Biobehavioral Processes	2025/01 ZRG1 BP-C (10) B

Public Sequence Contributions

NCBI SRA	BioProject: PRJNA1328365	*P. vivax* transcriptomes (MSP-1)	Ethiopia	40 samples
NCBI SRA	BioProject: PRJNA783000	*P. falicparum* CSP/ MSP-1 amplicon sequences	Ghana	90 samples
NCBI SRA	BioProject: PRJNA784582	*P. vivax* transcriptomes	Ethiopia	10 samples
GenBank	Accessions: OQ308838 to 53 OQ339140 to 49, OQ341668 to 74	*Clypeasteroida* mtDNA and rRNA	Taiwan	32 samples
NCBI GEO	Study: GSE252307	*D. rerio* MECP2 mutant/wild type transcriptomes	United States	20 samples

Trademarks

<*Tuple*> ® Service Mark	Reg: 8,113,385	Serial: 98-574,435	Classes: 35, 42	January 27, 2026

Funding

Jul 2026 - Dec 2026	*Empirical validation of computationally-designed antibodies through molecular AI models.* UNCC Faculty Research Grant, $5,000, Funded
Apr 2026 - Sep 2026	*Tuple Summer Internship Funding: Expanding AI-based Antibody Development through Diffusion and Structure-Aware Design* NCBiotech Industrial Internship Program, $4,000, Funded: 2026-IIP-0083
Apr 2025 - Sep 2025	*Tuple Summer Internship Funding: AI-Enabled Protein Design and Targeted Antibody Development* NCBiotech Industrial Internship Program, $4,000, Funded: 2025-IIP-0021
Oct 2024 - Sep 2026	*Empowering Advanced Alzheimer's Disease and Dementia Research Through Remote Patient Monitoring and Cloud-Based Wearable Devices* NIH|NIA SBIR Phase II, $2,523,349, Funded: 2R44AG072981-02A1
Aug 2024 - Jul 2025	*Augmented Preclinical Alzheimer's Disease Detection Through Wearable Health and Driving Behavior Data* NIH|NIA SBIR Phase I, $505,516, Funded: 1R43AG087803-01A1
Aug 2023 - Jul 2024	*Entomological Surveillance Dashboard, Harnessing Geographic Information Systems to Modernize Military Entomology* US DoD|Army, $150,000, Funded: W91YTZ23P0058
Aug. 2021 - Jul. 2023	*A Multifaceted Digital Health Application to Advance Alzheimer's Disease Patient Monitoring, Safety, Caregiving, and Research* NC SBIR Match, $75,000, Funded: 72494247 22-017
Aug 2021 - Jul 2023	*A Multifaceted Digital Health Application to Advance Alzheimer's Disease Patient Monitoring, Safety, Caregiving, and Research* NIH|NIA SBIR Phase I, $495,787, Funded: 1R43AG072981-01A1
Apr 2021 - Sep 2022	*Investigating the Utility of Digital Biomarkers with Wearable Devices and Artificial Intelligence in Alzheimer's Disease and other Neurological Disorders* NSF|I-Corps, $50,000, Funded: 2127407
May 2020 - Dec 2020	*Analytical Modeling of IoMT data to predict onset and progression of Alzheimer's* UNCC SDS Seed Grant, $20,000, Funded: 2020009
Jan 2019 - May 2019	*Azure Funding for Cloud Computing Course* Microsoft, $30,000, Funded: 5745151
Oct 2014 - Present	*Various Business Incentive Funds (BIF), Partner Investment Engine (PIE) Funds, GoFast Funding, and End Customer Investment Funds (ECIF)* Microsoft, >$750,000, Funded